FARM MANAGEMENT SYSTEM

Team ID: SWTID-2026-2906

Team Size: 4

Team Leader: MADHESH VARMA S

Team member: AAKASH G

Team member: MATHAN S

Team member: SARAVANAN M

ABSTRACT

The Farm Management System (FMS) is a Salesforce-based application designed to digitize and streamline agricultural operations. It manages data related to farms, crops, farmers, and buyers while providing real-time insights through reports and dashboards. The system helps farmers and managers monitor agricultural activities efficiently and make informed decisions.

S.NO	CONTENTS
1	INTRODUCTION
2	IDEATION PHASE
3	REQUIREMENT ANALYSIS PHASE
4	PROJECT DESIGN PHASE
5	PROJECT PLANNING AND SCHEDULING
6	PROJECT DEVELOPMENT PHASE
7	FUNCTIONAL AND PERFOMANCE TESTING

1.INTRODUCTION

The Farm Management System (FMS) is a Salesforce-based application designed to digitize and streamline agricultural operations. It manages data related to farms, crops, farmers, and buyers while providing real-time insights through reports and dashboards. The system helps farmers and managers monitor agricultural activities efficiently and make informed decisions.

2. IDENTATION PHASE

The main purpose of this project is to:

- Digitize agricultural data management
- Improve data accuracy and accessibility
- Automate record handling
- Enable real-time monitoring
- Support data-driven farming decisions
- Enhance productivity and transparency

3. Empathy Map Canvas

Target Users

- Farmers
- Farm Managers
- Buyers
- System Administrators

Empathy Analysis

- Think & Feel: Need accurate and timely farm data
- Hear: Feedback from agricultural officers
- See: Registers, paper files, spreadsheets
- Say & Do: Enter records manually
- Pain: Data loss, errors, delays
- Gain: Automation, fast reporting

3.1 Brainstorming

Brainstormed Ideas

- Manual farm records
- Excel-based management
- Web-based system
- Mobile farm app
- Salesforce-based FMS

Selected Idea

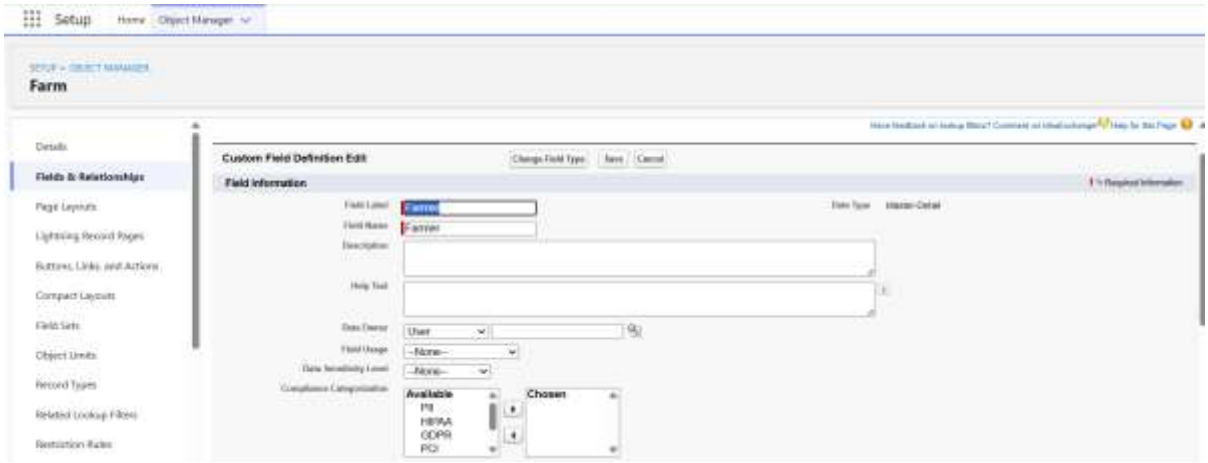
Salesforce-based Farm Management System.

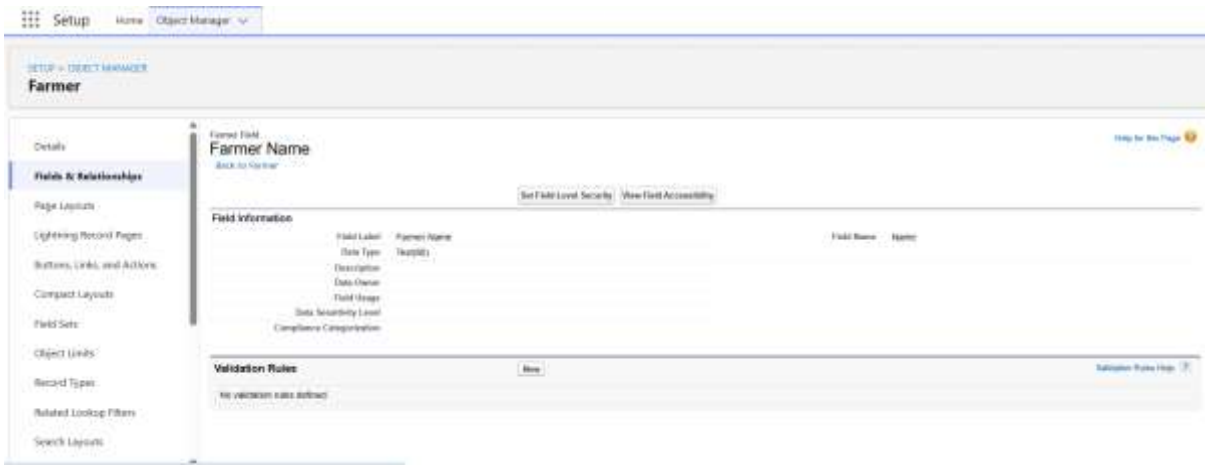
3.2 Solution Requirements

Functional Requirements

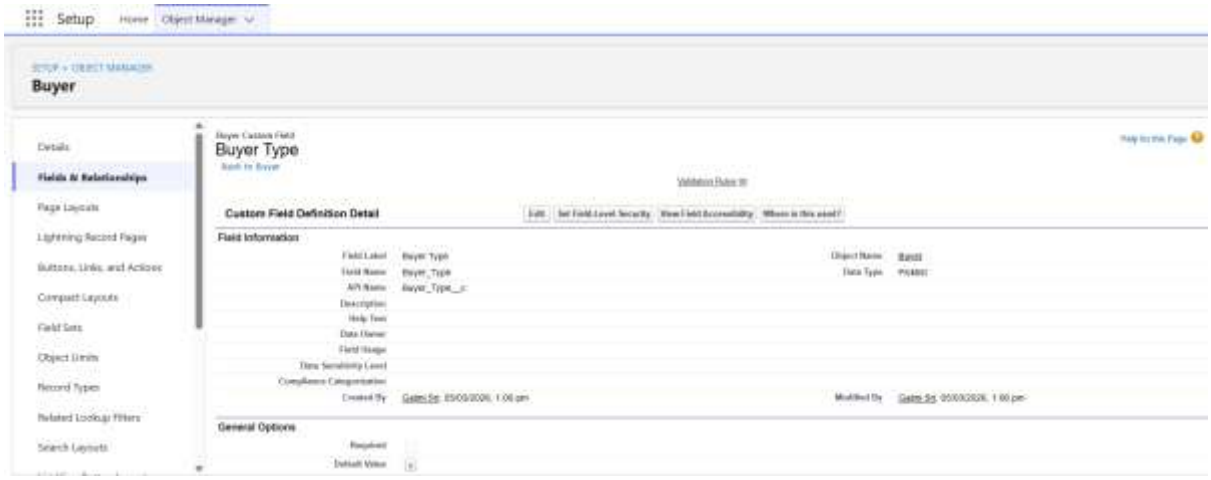
FR No	Requirement
FR-1	User Authentication
FR-2	Farm, Crop, Farmer, Buyer Management
FR-3	Data Entry and Validation
FR-4	Relationship Mapping
FR-5	Report Generation
FR-6	Dashboard Visualization
FR-7	User Role Management

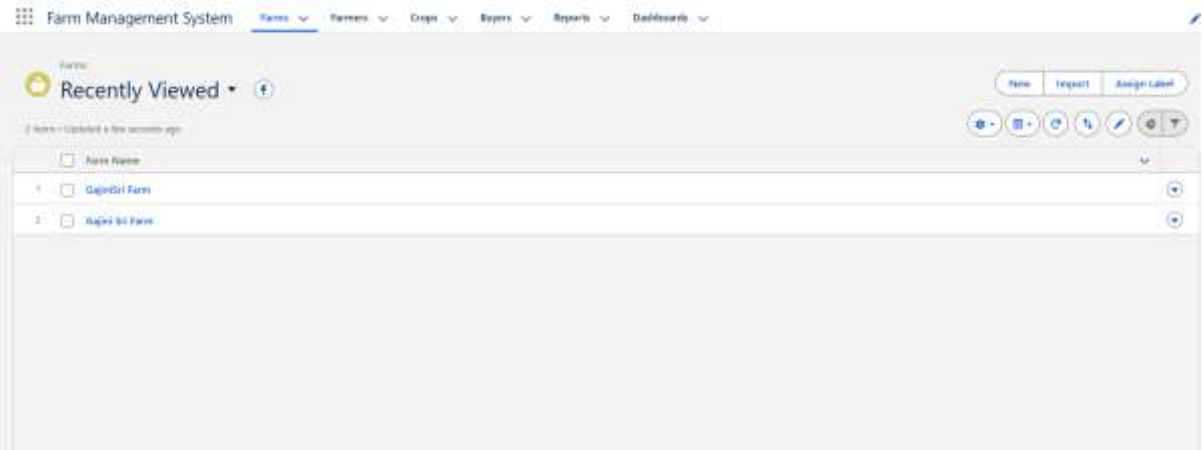
Non-Functional Requirements

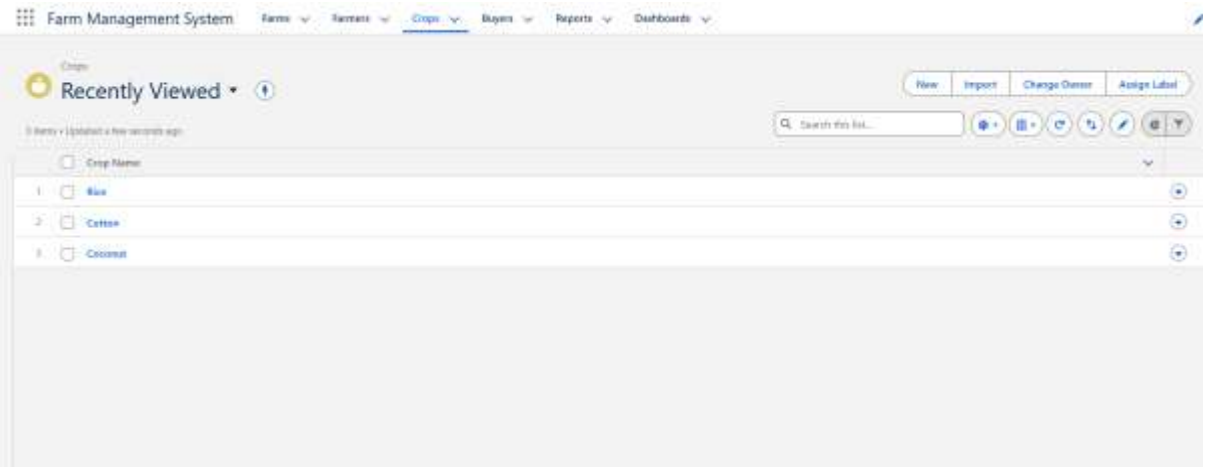

Setup
Home
Object Manager
Farm
Details
Fields & Relationships
Page Layouts
Lightning Record Pages
Buttons, Links, and Actions
Compact Layouts
Field Sets
Object Limits
Record Types
Related Lookup Filters
Restriction Rules
Custom Field Definition Edit
Change Field Type
Save
Cancel
Field Information
Field Label
Farmer
Field Name
Farmer
Description
Help Text
Data Owner
User
Field Usage
--None--
Data Sensitivity Level
--None--
Compliance Categorization
Available
PII
HIPAA
GDPR
PCI
Chosen

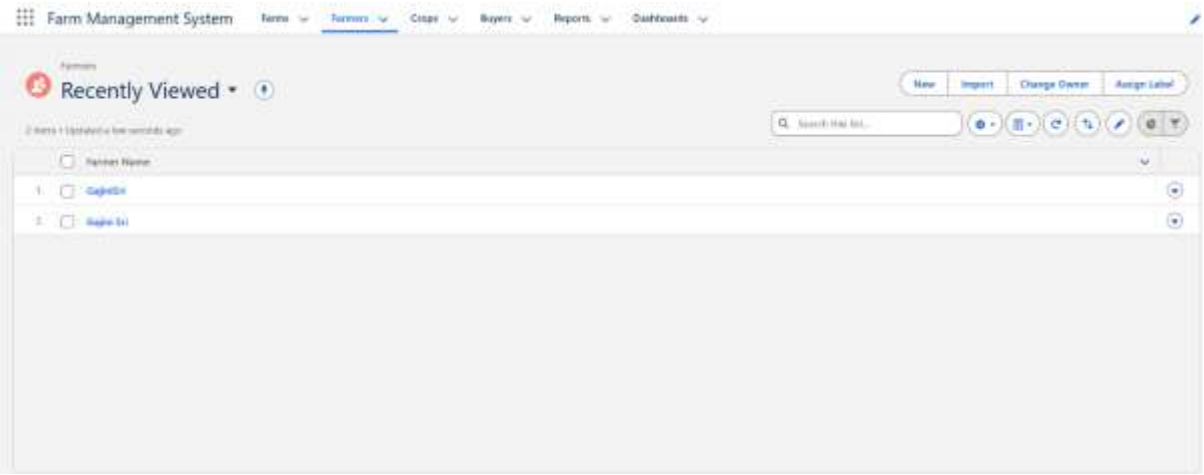

Setup
Home
Object Manager
Farmer
Details
Fields & Relationships
Page Layouts
Lightning Record Pages
Buttons, Links, and Actions
Compact Layouts
Field Sets
Object Limits
Record Types
Related Lookup Filters
Search Layouts
Farmer Name
Field Information
Field Label
Farmer Name
Field Name
Name
Data Type
Text(80)
Validation Rules
New

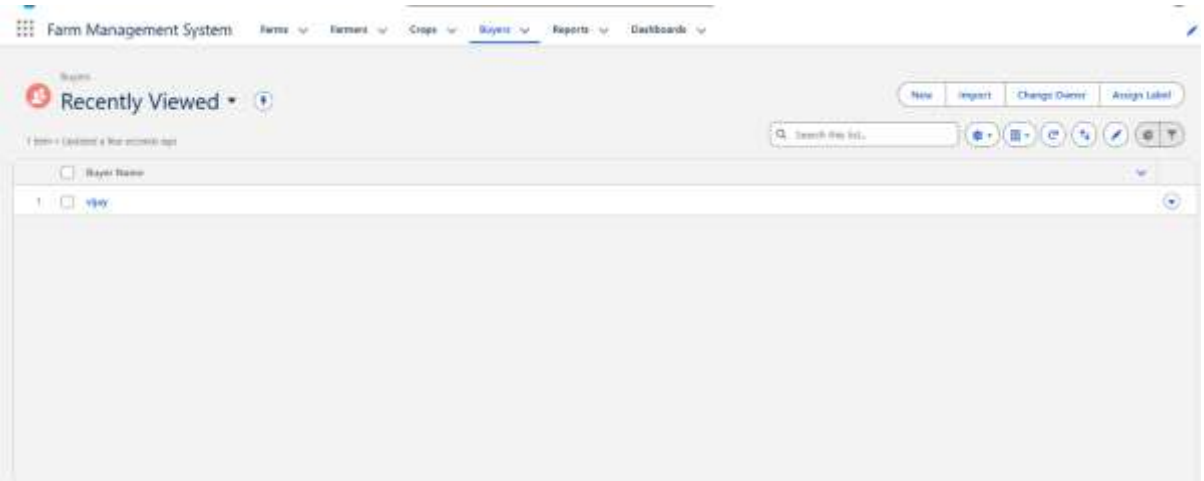
Setup
Home
Object Manager
Crop
Details
Fields & Relationships
Page Layouts
Lightning Record Pages
Buttons, Links, and Actions
Compact Layouts
Field Sets
Object Limits
Record Types
Related Lookup Filters
Search Layouts
Crop Name
Custom Field Definition Detail
Edit
Set Field-Level Security
View Field Accessibility
Where is this used?
Field Information
Field Label
Crop Name
Field Name
Crop_Name
API Name
Crop_Name__c
Object Name
Crop
Data Type
Text
General Options
Required
Unique

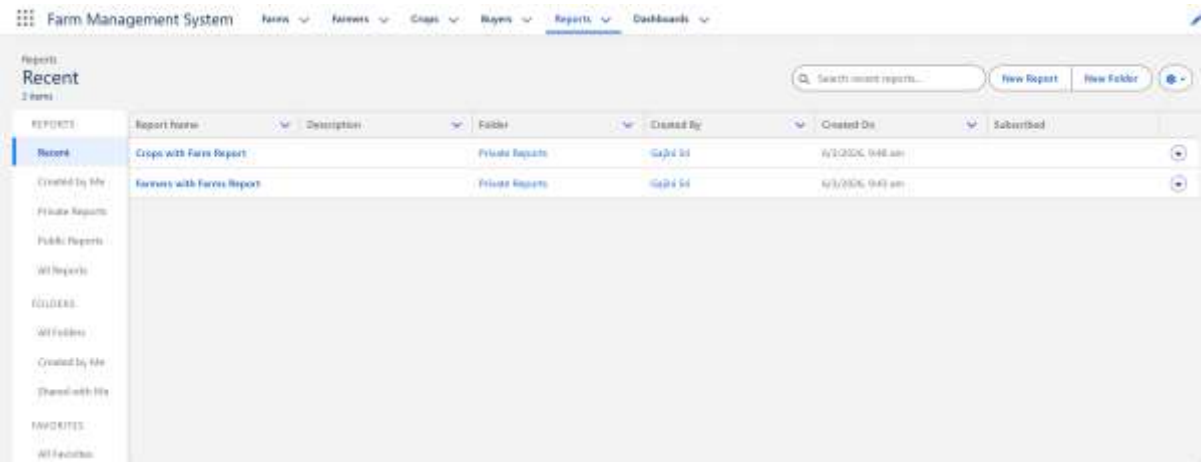

Setup
Home
Object Manager
Buyer
Details
Fields & Relationships
Page Layouts
Lightning Record Pages
Buttons, Links, and Actions
Compact Layouts
Field Sets
Object Limits
Record Types
Related Lookup Filters
Search Layouts
Buyer Type
Custom Field Definition Detail
Field Information
Buyer Type
Buyer_Type
Buyer_Type__c
General Options

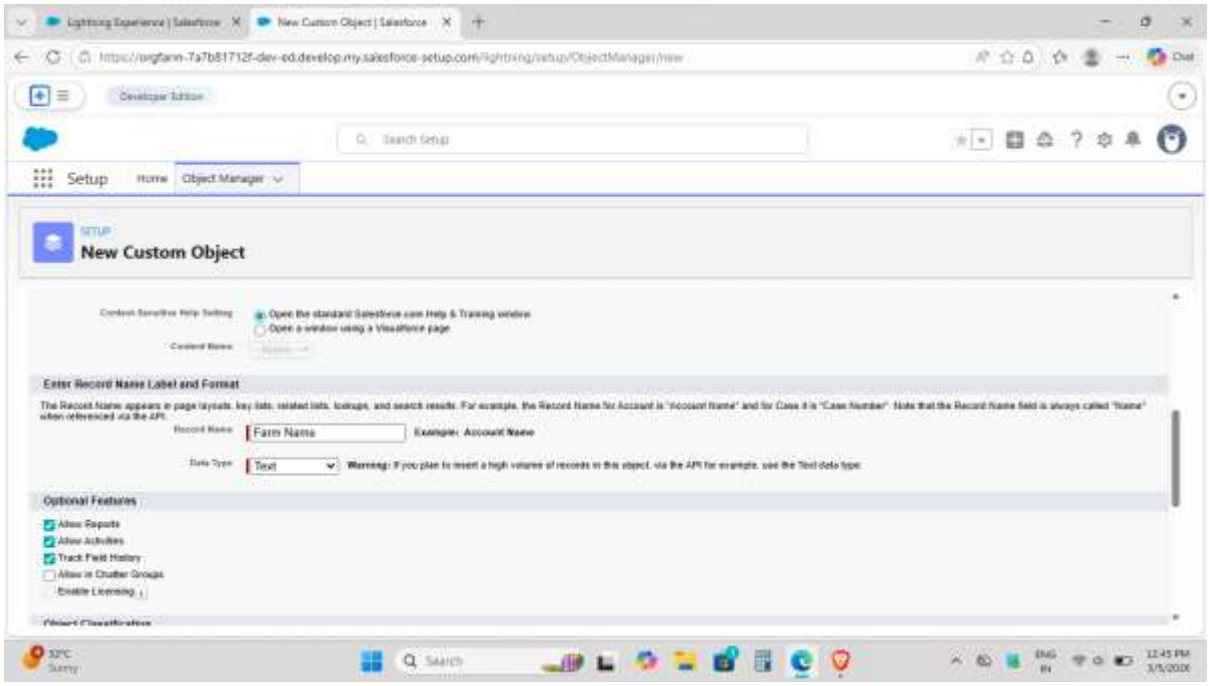

Farm Management System
Farms
Farmers
Crops
Buyers
Reports
Dashboards
Recently Viewed
New
Import
Assign Label
Farm Name
Gajendra Farm
Rajesh Sri Farm

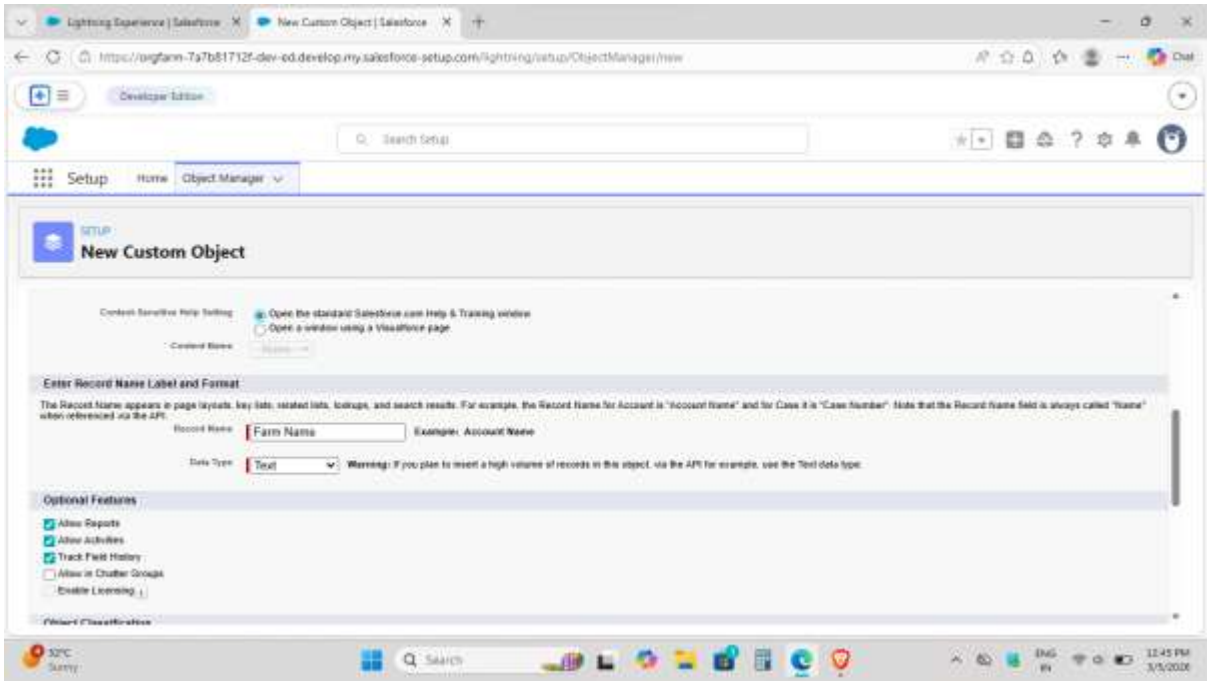

Farm Management System
Farms
Farmers
Crops
Buyers
Reports
Dashboards
Recently Viewed
New
Import
Change Owner
Assign Label
Crop Name
Rice
Cotton
Coconut

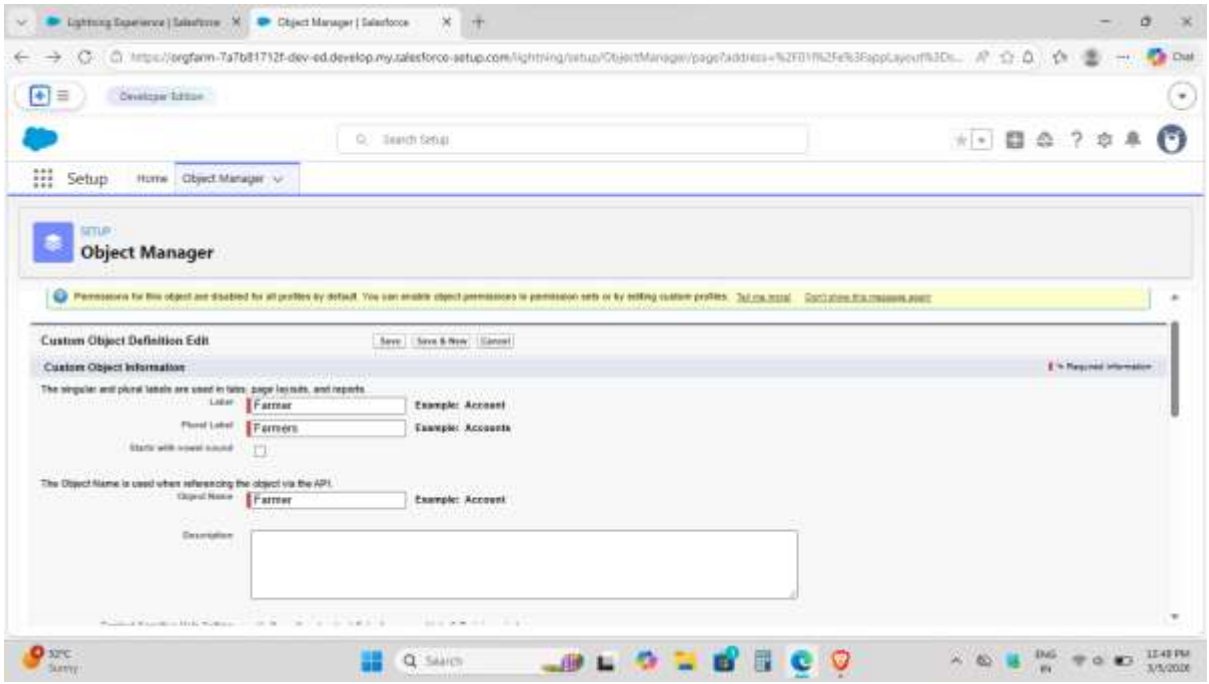

Farm Management System
Farms
Farmers
Crops
Buyers
Reports
Dashboards
Recently Viewed
New
Import
Change Owner
Assign Label
Farmer Name

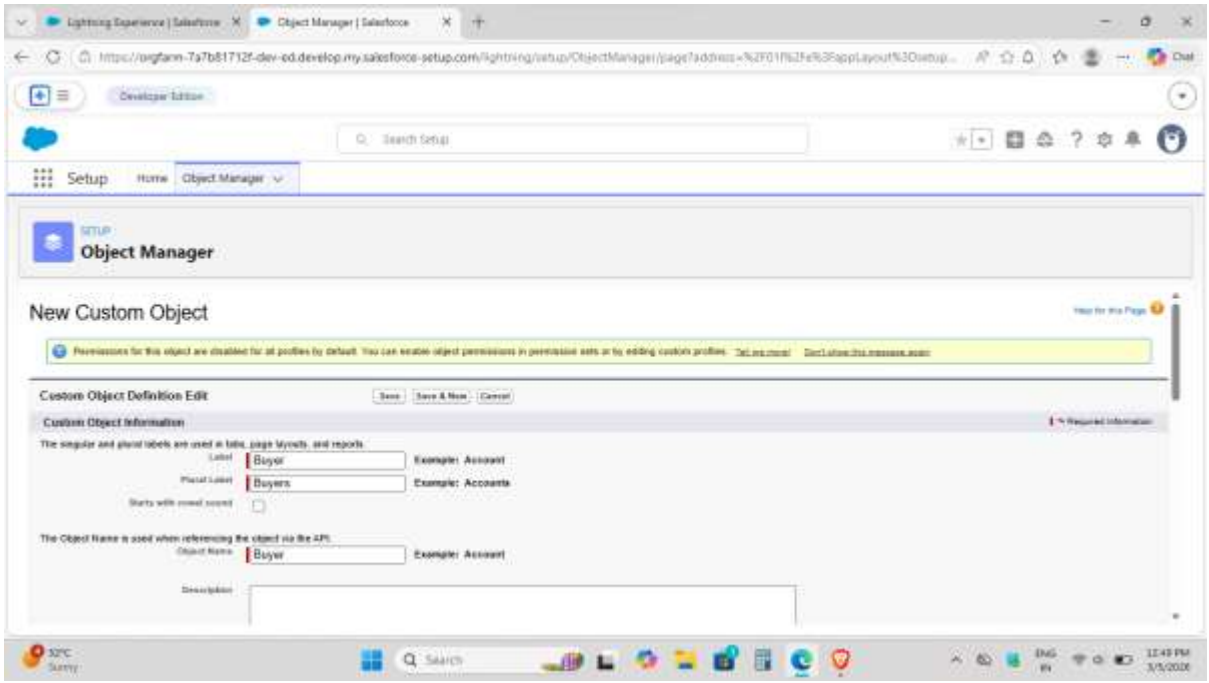

Farm Management System
Farms
Farmers
Crops
Buyers
Reports
Dashboards
Recently Viewed
New
Import
Change Owner
Assign Label
Buyer Name
vijay

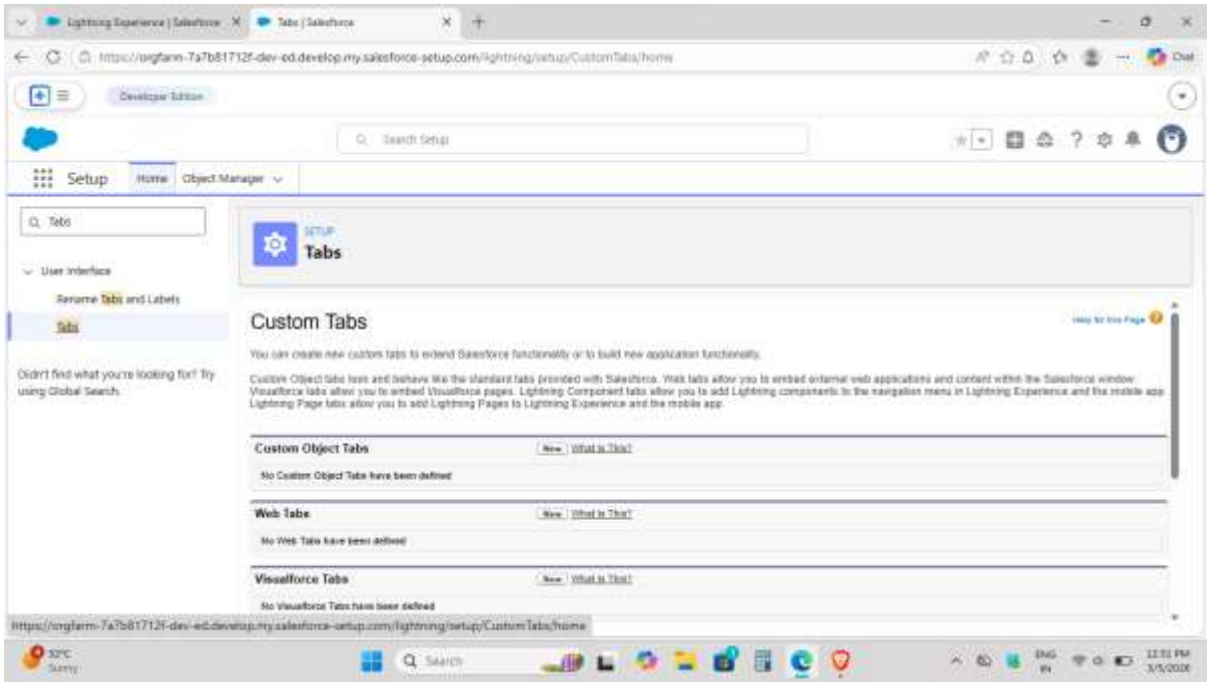

Farm Management System
Farms
Farmers
Crops
Buyers
Reports
Dashboards
Recent
New Report
New Folder
REPORTS
Recent
Report Name
Description
Folder
Created By
Created On
Subscribed
Crops with Farm Report
Private Reports
Farmers with Farms Report
Private Reports

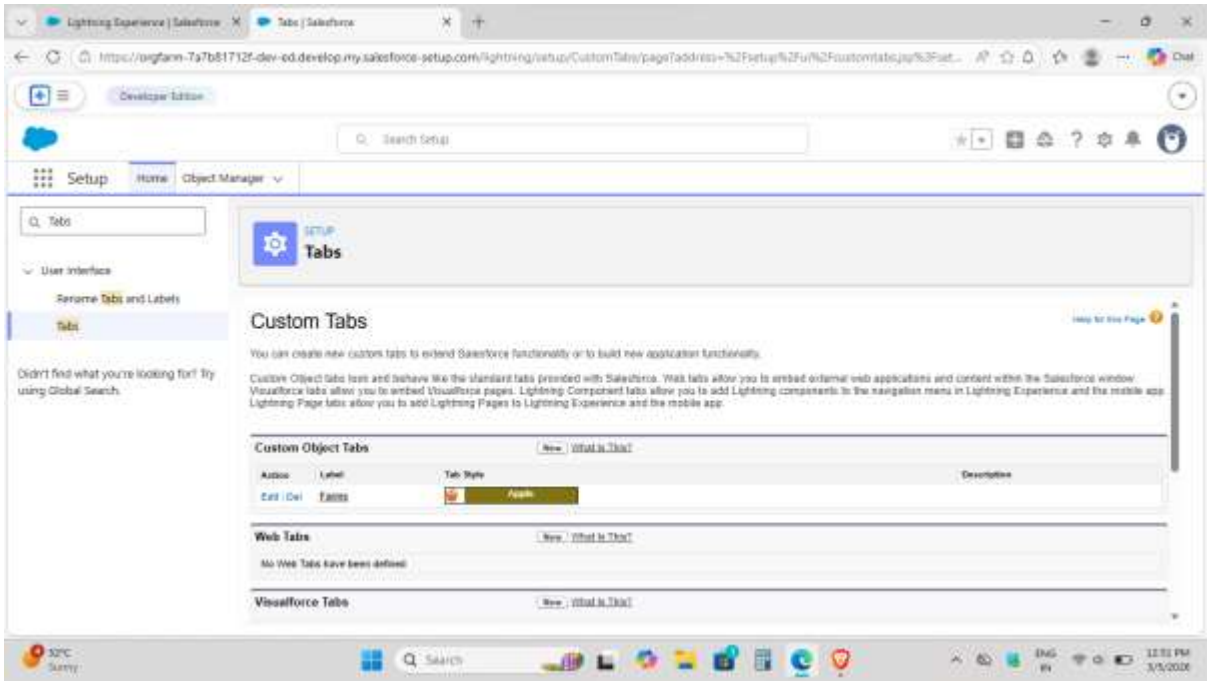

Lightning Experience | Salesforce
New Custom Object | Salesforce
Developer Edition
Search Setup
Setup
Home
Object Manager
SETUP
New Custom Object
Enter Record Name Label and Format
Farm Name
Text
Optional Features
Allow Reports
Allow Activities
Track Field History


Lightning Experience | Salesforce
New Custom Object | Salesforce
Developer Edition
Search Setup
Setup
Home
Object Manager
SETUP
New Custom Object
Open the standard Salesforce.com Help & Training window
Open a window using a Visualforce page
Enter Record Name Label and Format
Farm Name
Example: Account Name
Text
Optional Features
Allow Reports
Allow Activities
Track Field History
Allow in Chatter Groups
Enable Licensing


Lightning Experience | Salesforce
Object Manager | Salesforce
Developer Edition
Search Setup
Setup
Home
Object Manager
SETUP
Object Manager
Custom Object Definition Edit
Save
Save & New
Cancel
Custom Object Information
Farmer
Example: Account
Farmers
Example: Accounts
Farmer
Example: Account


Lightning Experience | Salesforce
Object Manager | Salesforce
Developer Edition
Search Setup
Setup
Home
Object Manager
SETUP
Object Manager
New Custom Object
Custom Object Definition Edit
Save
Save & New
Cancel
Custom Object Information
Buyer
Example: Account
Buyers
Example: Accounts
Buyer
Example: Account


Lightning Experience | Salesforce
Tabs | Salesforce
Developer Edition
Search Setup
Setup
Home
Object Manager
Tabs
User Interface
Rename Tabs and Labels
Tabs
Didn't find what you're looking for? Try using Global Search.
SETUP
Tabs
Custom Tabs
Custom Object Tabs
New
What is This?
No Custom Object Tabs have been defined
Web Tabs
No Web Tabs have been defined
Visualforce Tabs
No Visualforce Tabs have been defined



4. REQUIREMENT ANALYSIS PHASE

4.1 Problem–Solution Fit

The FMS replaces manual farming records with automated data management. Cloud storage prevents data loss, dashboards improve analysis, and real-time reports eliminate delays. This ensures accurate and timely agricultural insights.

4.2 Solution Architecture

Architecture Layers

1. Presentation Layer – Salesforce Lightning UI
2. Application Layer – Validation Rules, Relationships
3. Data Layer – Custom Objects
4. Analytics Layer – Reports and Dashboards

5. PROJECT PLANNING & SCHEDULING

5.1 Project Planning

- Created Salesforce Developer Org
- Developed custom objects
- Configured tabs and Lightning App
- Created fields and relationships
- Designed profiles and roles
- Created users and sharing rules
- Developed reports and dashboards
- Conducted testing and reviews

Product Backlog, Sprint Schedule, and Estimation

Use the below template to create product backlog and sprint schedule

6. Project Development Phase

Project Description:

The Farm Management System (FMS) in Salesforce is a comprehensive solution designed to digitize and streamline agricultural operations. This application empowers farmers, farm managers, and stakeholders by managing data related to crops, Farm, Farmer, and buyers, while providing actionable insights through reports, and dashboards.

Milestone 1-Salesforce:

What Is Salesforce?

Salesforce is your customer success platform, designed to help you sell, service, market, analyze, and connect with your customers.

Salesforce has everything you need to run your business from anywhere. Using standard products and features, you can manage relationships with prospects and customers, collaborate and engage with employees and partners, and store your data securely in the cloud.

7. FUNCTIONAL & PERFORMANCE TESTING

During the testing phase, screenshots were captured from the Salesforce interface to verify system functionality. The following were tested:

- Creation of Farm, Crop, Farmer, and Buyer records
- Validation of input fields
- Relationship and dependency checks
- Profile and role access testing
- Sharing rule verification
- Report and dashboard generation

Performance testing ensured smooth system response and stability during multiple user operations.

8. ADVANTAGES & DISADVANTAGES

Advantages

- Centralized data system
- Reduced paperwork
- Improved accuracy
- Better monitoring
- Real-time analytics

Disadvantages

- Requires internet connection
- Salesforce license dependency
- Limited offline access

9. Conclusion

The Farm Management System successfully digitizes agricultural data management using Salesforce. It improves efficiency, reduces errors, and enables farmers and managers to make informed decisions through automation and analytics.

Thank You